



Boundary conditions in dissipative particle dynamics

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Abstract

We study a continuum model for simulating solid boundaries of a DPD fluid. A layer of frozen DPD particles is placed on the wall boundary and a continuum limit is taken. This produces effective dissipative and random forces. In addition, a reflection at the wall must be specified in order to confine the fluid. Three different wall reflection laws are studied and their effectiveness in producing stick boundary conditions and correct temperature distributions is assessed by simulation of a planar Couette flow. © 1999 Elsevier Science B.V. All rights reserved.

Dissipative Particle Dynamics (DPD) was introduced originally as an attempt to exploit the computational advantages of lattice-gas methods in computational fluid dynamics and at the same time avoid the problems of being defined in a lattice [1,2].

As in any other method in computational fluid dynamics, the issue of boundary conditions has to be addressed in DPD. Until now, three methods have been considered in the treatment of boundary conditions in DPD. In the first method, in order to simulate a shear flow as that generated in a Couette geometry, the Lees–Edward technique has been used [3,4], which is a way of avoiding the modelization of physical boundaries. A second alternative is to “freeze” some portions of the fluid described with DPD. For example, in the modelling of colloidal suspensions a fictitious sphere is placed inside the fluid and all the DPD particles within the sphere are obliged to move as a rigid solid [1,3]. The same method can be used for modelization of solid walls confining the DPD fluid. Finally, another method developed by the authors exploits this latter idea of freezing particles in a

generic model for treatment of the boundaries between solid objects and DPD particles [5]. We considered a layer of DPD particles that are stuck onto the solid; by taking a continuum limit of this layer an effective force due to the solid onto the DPD fluid particles was derived.

In this latter model, the effective forces are not sufficient for keeping the fluid confined and one has to specify what happens when a DPD particle crosses the line that defines the position of the wall. We have investigated three different possibilities [5]:

- (1) Specular reflections such that the parallel component of the momentum of the particle is conserved and the normal component is reversed.
- (2) Maxwellian reflections where the particles are introduced back into the system according to a Maxwellian distribution of velocities centered at the velocity of the wall.
- (3) Bounce back reflections in which both components of the velocity are reversed.

We have computed the velocity profile in 2D shear flow produced by two parallel walls at $y = 0$ and $y = h$ with the upper wall moving at constant velocity V . A crucial dimensionless parameter in the model

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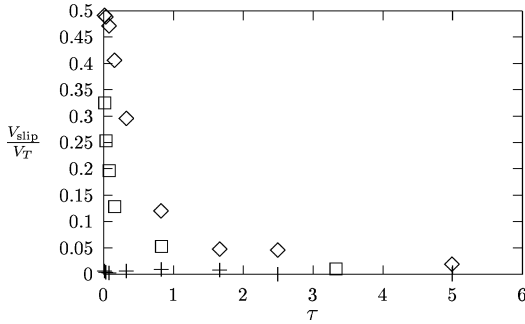


Fig. 1. Slip velocity at the wall for the specular (diamonds), Maxwellian (squares), and bounce back (crosses) reflections versus τ .

is the dimensionless friction coefficient defined by $\tau = \gamma \lambda / d v_T$ where γ is the friction coefficient, λ is the average distance between particles, d is the spatial dimension (equal to 2) and $v_T = \sqrt{k_B T / m}$ is the thermal velocity [5]. For high values of τ , the above three wall reflection possibilities produce a linear velocity profile along the channel and the fluid and wall velocities match at the boundaries, i.e. there is no slip.

For small τ slip appears in specular and Maxwellian reflections and only bounce back produces stick boundary conditions for any values of τ . This is shown in Fig. 1 where the slip velocity is plotted against τ . In principle, bounce back would be the preferred selection. However, we show below that at small τ , even though bounce back produces correct velocity profiles, it does show some artifacts concerning the temperature profile.

We define the quantities T^x and T^y by

$$T^x = \sum_i m_i v_{x_i}^2, \quad T^y = \sum_i m_i v_{y_i}^2, \quad (1)$$

where the velocities are taken with respect to the average mean flow. We see that the temperature is given by

$$T = \frac{1}{2} (T^x + T^y)$$

and we loosely call T^x and T^y the two components of the temperature. We observe in the simulation that the two components of the temperature are constant along the channel. This is done by binding the channel and counting in the sums in the definitions (1) only the particles that are in each bin. In Fig. 2 we plot the two components of the temperature (1) for the

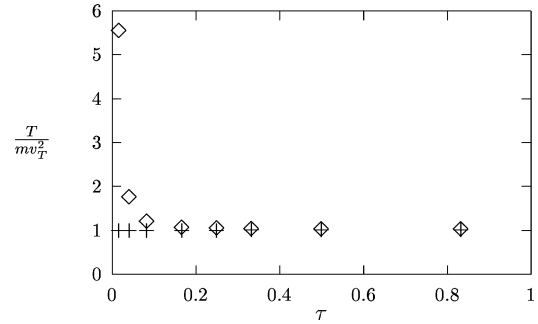


Fig. 2. T^x (diamonds) and T^y (crosses) versus τ when $T = 1$.

case of bounce back boundary conditions. For high values of τ , T^x and T^y coincide and their sum is equal to half the imposed temperature. As τ decreases, the function T^x increases while T^y remains constant at the system temperature. This is anomalous because we expect that both components of the temperature should be equal. Actually, for the case of specular reflections there is no significant discrepancy between T^x and T^y at any τ (there is a slight discrepancy for Maxwellian reflections, although not so pronounced as for the bounce back reflections).

Although bounce back boundary conditions give the correct velocity profiles for all τ , for small τ they produce higher temperature than the one imposed. This effect is not related to the finite time step used in the discretization of the stochastic equations governing DPD [6]. The origin of this effect is due to the great dispersion in the component along the flow of the velocity. In turn, this is due to the fact that for small τ the interaction between fluid particles is small. For $\tau = 0$, the DPD particles move in straight lines and the bounce back boundary condition accelerates the particles without bound in each collision with the wall. The dispersion of the x component of the velocity, therefore, increases without bound. At small but finite τ the friction saturates the dispersion to a finite value.

The conclusion is that all the wall reflecting laws we have studied provide good results for large τ in the sense that all three produce stick boundary conditions. However, for small values of τ the three methods produce spurious results. Specular and Maxwellian reflections lead to slip on the walls whereas bounce back reflections show an anomalous temperature behavior.

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